FOP – Lab – Task 1.2

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Problem:

Create a simple self-checkout system. Prompt for the prices and quantities of three items. Calculate the subtotal of the items. Then calculate the tax using a tax rate of 5.5%. Print out the line of items with the quantity and total, and then print out the subtotal, tax amount, and total.

Example Output

Enter the price of item 1: 25

Enter the quantity of item 1: 2
Enter the price of item 2: 10
Enter the quantity of item 2: 1
Enter the price of item 3: 4
Enter the quantity of item 3: 1
Subtotal: $64.00 Tax: $3.52
Total: $67.52

Constraints

- Keep the input, processing, and output parts of your program separately. Collect the input, then do the math operations and string building, and then print out the output.
- Be sure you explicitly convert input to numerical data before doing any calculations.

Challenges

- Revise the program to ensure that prices and quantities are entered as numeric values. Don't allow the user to proceed if the value entered is not numeric.
- Alter the program so that an indeterminate number of items can be entered. The tax and total are computed when there are no more items to be entered

Algorithm:

Step 1:

Program starts and displays "Bill Application" and "Enter Numeric Values in all entries."

Step 2:

Declare variables aq, ap, bq, bp, cq, cp, aa, ba, ca, st, tax, and T.

Step 3:

Taking Inputs of Quantities a,b,c and Prices of a, b, and c.

Step 4:

Check whether all inputs are numerical value or not. If all inputs are numeric then proceed the program to step 5. Else display "Invalid Inputs" and go back to step 3 to take inputs again.

Step 5:

Store these inputs in variables aq, bq, cq, ap, bp, and cp respectively.

Step 6:

Multiply aq and ap and store it in aa which makes our total amount for one product.

Multiply bq and bp and store it in ba which makes our total amount for one product.

Multiply cq and cp and store it in ca which makes our total amount for one product.

Step 7:

Compute subtotal by adding aa, ba, and ca. Stored subtotal in variable st.

Compute tax on subtotal by this formula:

tax = (5.5/100) * subtotal, store tax in variable named tax.

Compute total amount user has to pay by adding subtotal(st) and Tax(tax). Store it in T variable.

Step 8:

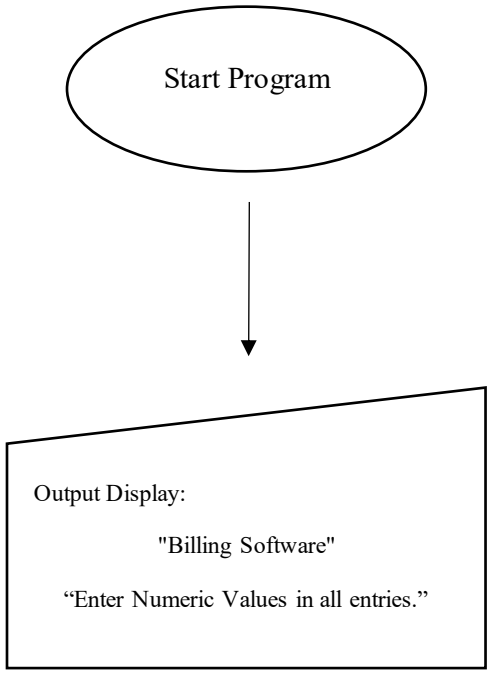
Display the final bill in an order by displaying the rate of a single item, its quantity, and total price of this item retrieve these values from respective variables

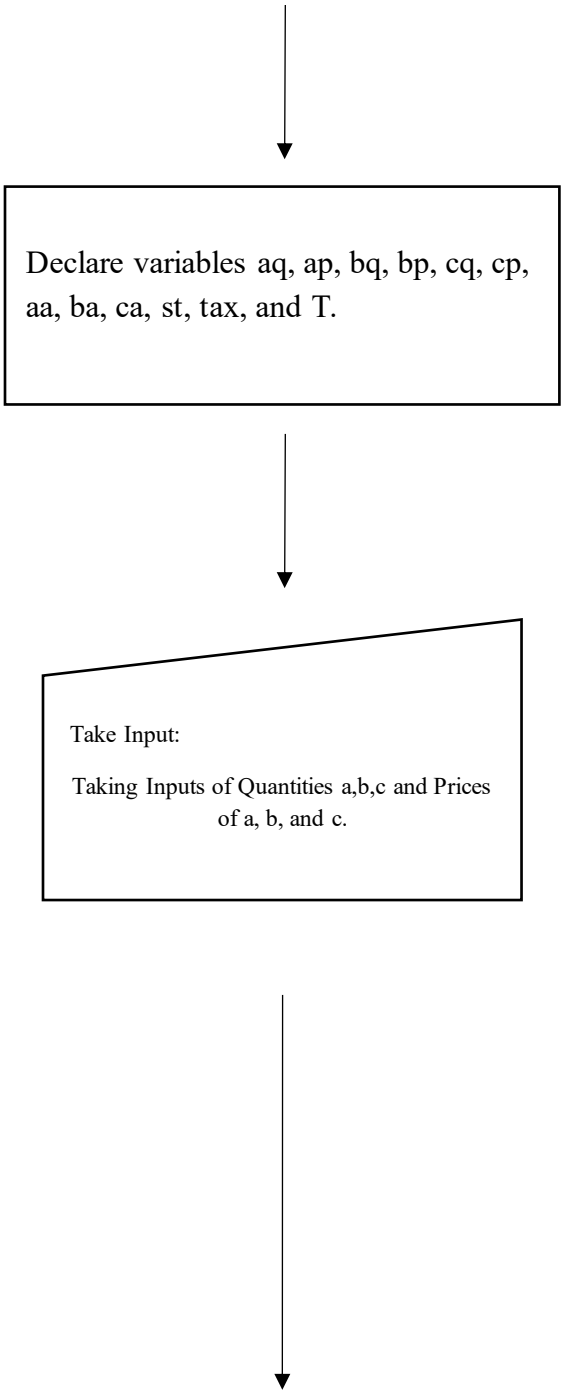
At the end display subtotal from variable st, tax form variable tax, and Grand total from variable T.

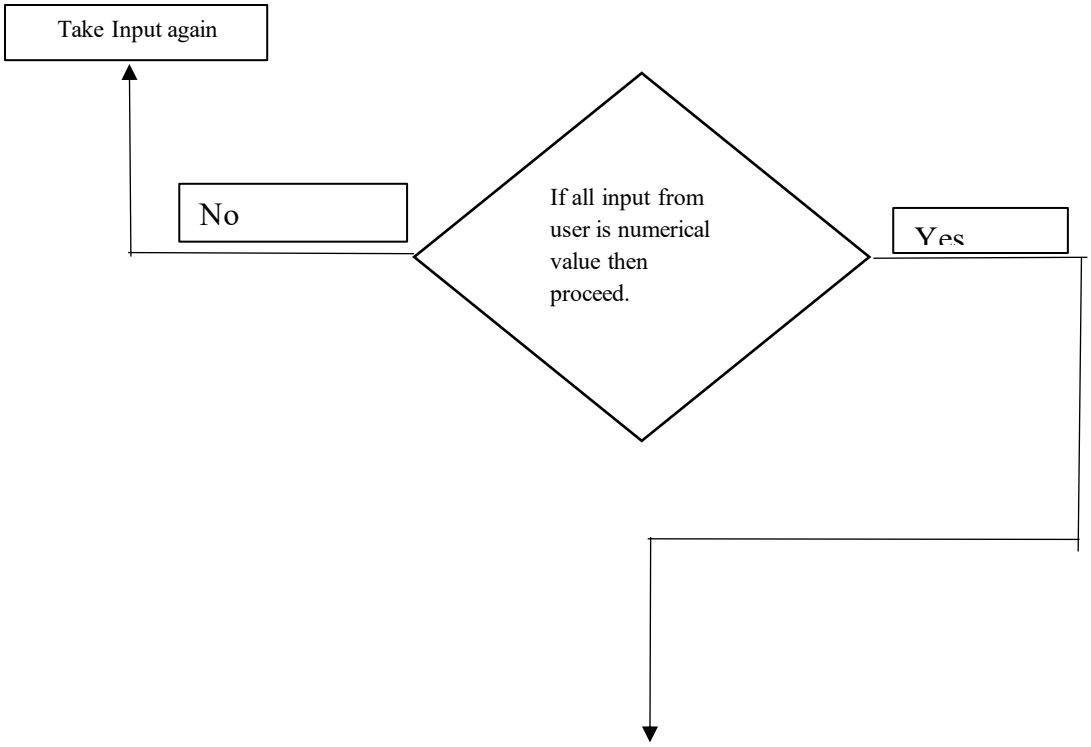
Step 9:

Stop the Program.

Flowchart:




Declare variables aq, ap, bq, bp, cq, cp, aa, ba, ca, st, tax, and T.
Take Input:
Taking Inputs of Quantities a,b,c and Prices of a, b, and c.



1) Store these inputs in variables aq, bq, cq, ap, bp, and cp respectively.

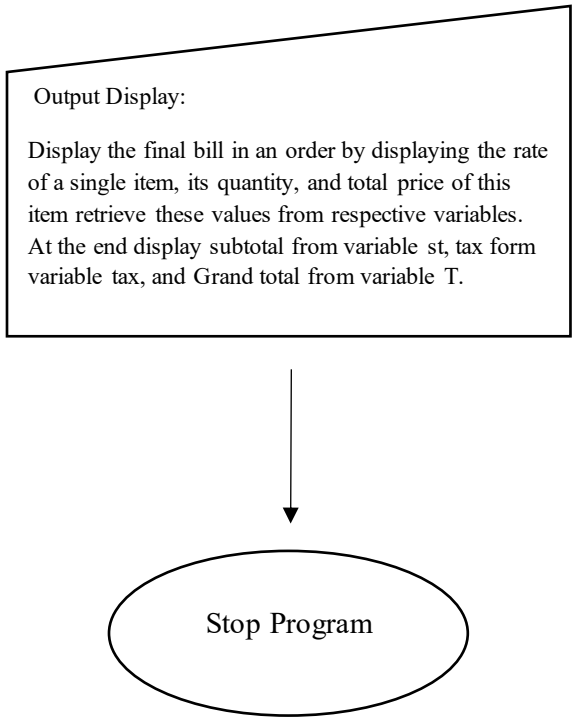
2) Multiply aq and ap and store it in aa which makes our total amount for one product. Multiply bq and bp and store it in ba which makes our total amount for one product. Multiply cq and cp and store it in ca which makes our total amount for one product.

3) Compute subtotal by adding aa, ba, and ca. Stored subtotal in variable st.

4) Compute tax on subtotal by this formula:

tax = (5.5/100) * subtotal, store tax in variable named tax.

5) Compute total amount user has to pay by adding subtotal(st) and Tax(tax). Store it in T variable.


Output Display:
Display the final bill in an order by displaying the rate of a single item, its quantity, and total price of this item retrieve these values from respective variables. At the end display subtotal from variable st, tax form variable tax, and Grand total from variable T.
Stop Program